

analysis

May 6, 2026

1 LLM-Augmented Stock Return Prediction — Analysis

Course: Artificial Intelligence and Machine Learning, Copenhagen Business School, Summer Term 2026

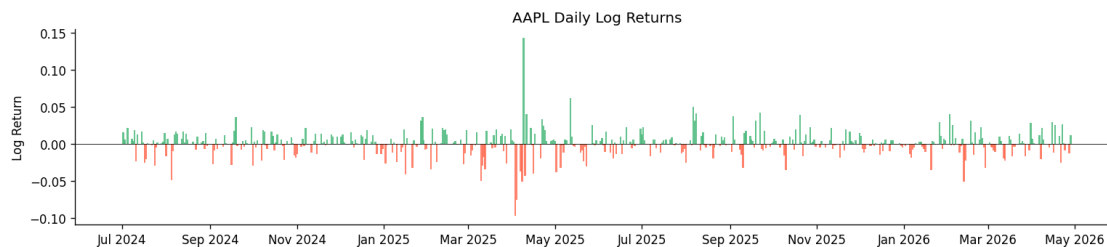
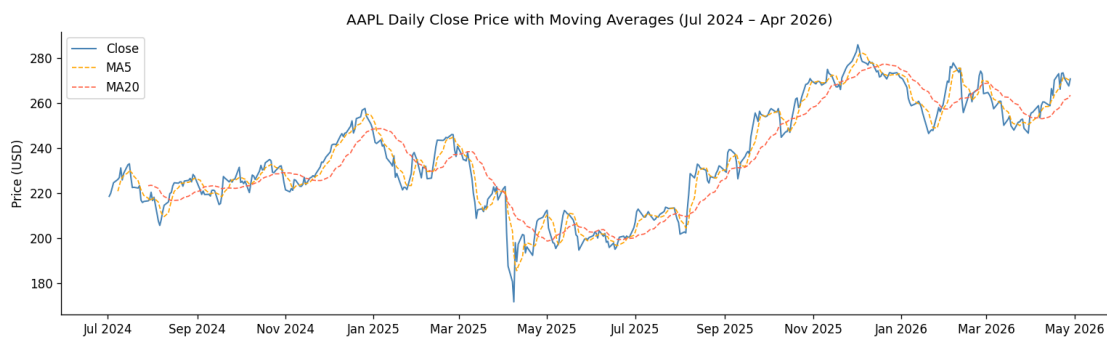
Research question: ...

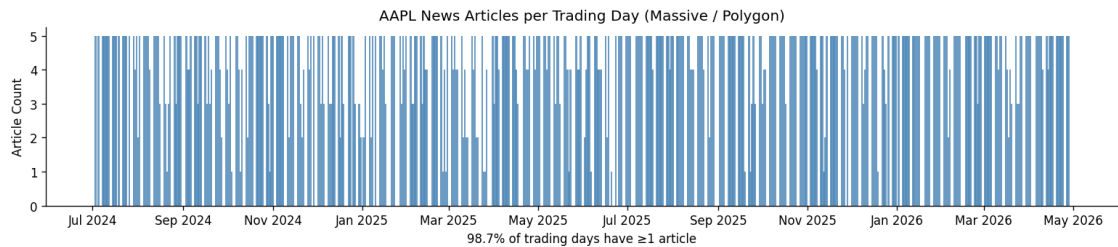
Condition	Description	Method
B	LSTM regressor	LSTM on OHLCV + technicals → log return → direction
B1	LSTM classifier (single-day)	LSTM with BCELoss, seq_len=1 → direction
B2	LSTM classifier (sequence)	LSTM with BCELoss, seq_len=20 → direction
L1	LLM — price only	GPT-5-nano on last 10 days of price data → up/down
L2	LLM — news only	GPT-5-nano on top-5 article summaries → up/down
L3	LLM — price + news	GPT-5-nano on price + news → up/down
X1	XGB regressor (single-day)	XGBoost on current-day features → log return → direction
X2	XGB regressor (20-day lags)	XGBoost on flattened 20-day features → log return → direction
X3	XGB classifier (single-day)	XGBoost on current-day features → direction
X4	XGB classifier (20-day lags)	XGBoost on flattened 20-day features → direction

Condition	Description	Method
XN	XGB classifier + news	X3 + PCA-reduced LLM report embeddings → direction

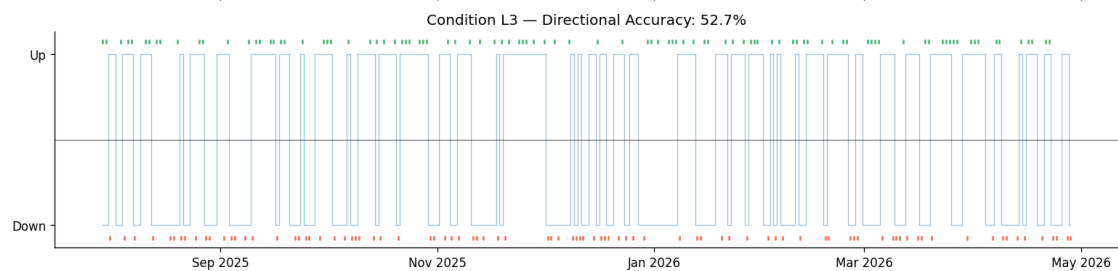
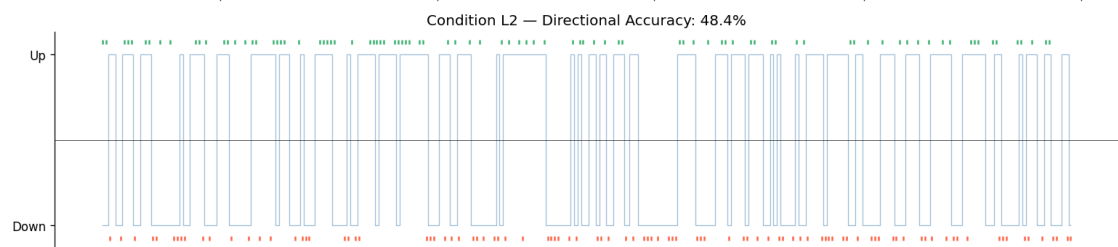
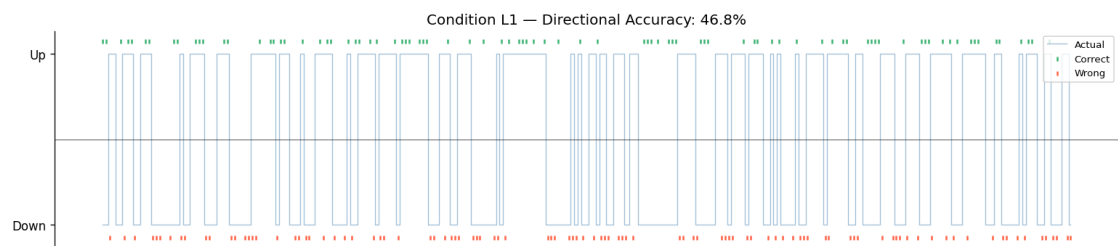
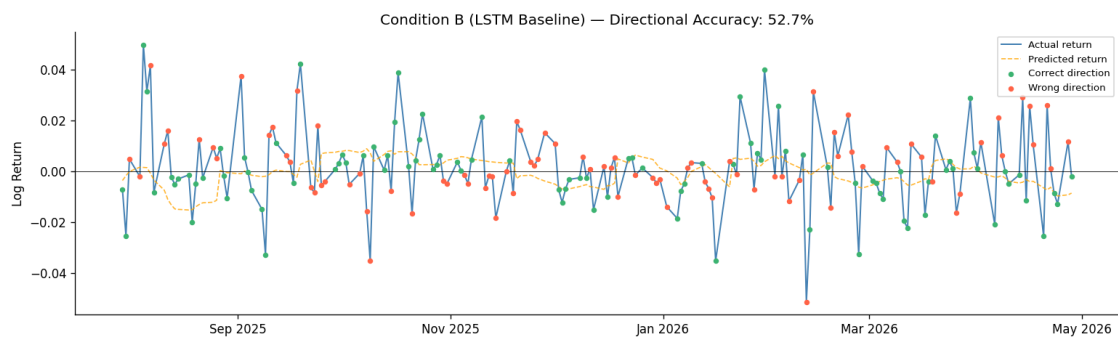
pred_B : 188 days
 pred_B1 : 188 days
 pred_B2 : 188 days
 pred_L1 : 188 days
 pred_L2 : 188 days
 pred_L3 : 188 days
 pred_X1 : 188 days
 pred_X2 : 188 days
 pred_X3 : 188 days
 pred_X4 : 188 days
 pred_XN : 188 days

1.1 Section 1 — Data Overview





1.2 Section 2 — Walk-Forward Prediction Charts



1.3 Section 2b — XGBoost & LSTM Classifier Predictions



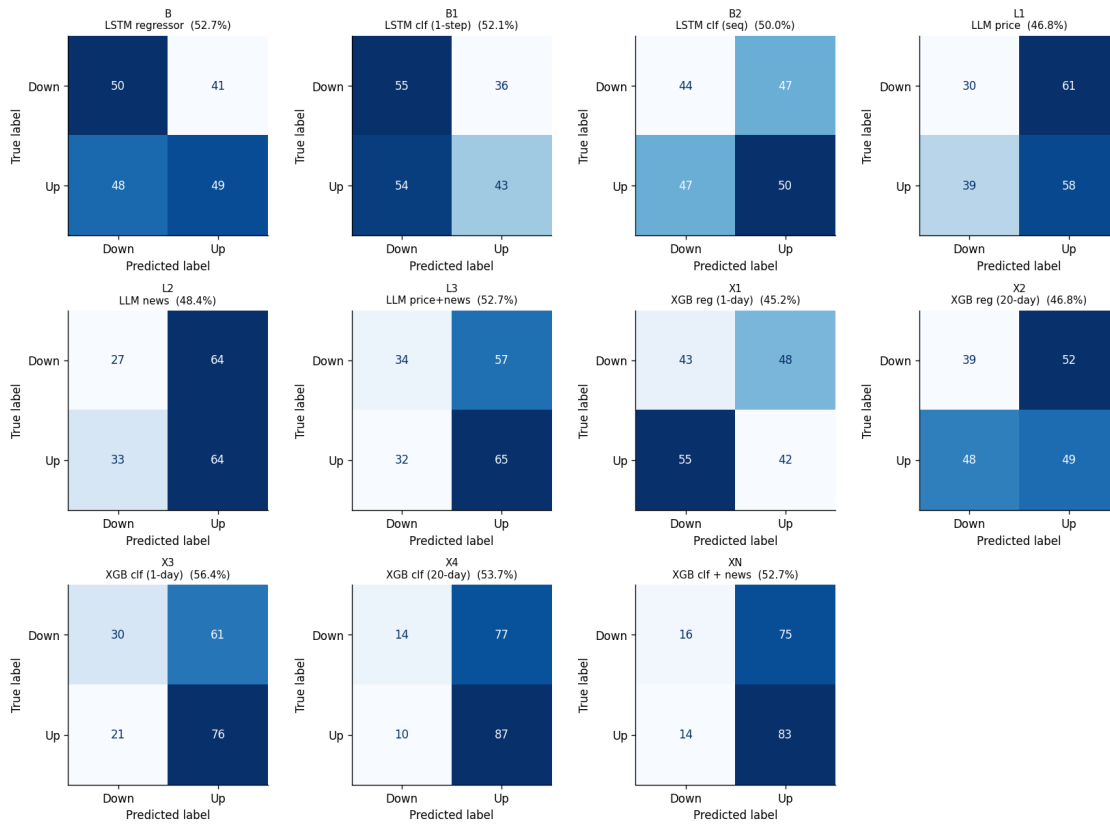
1.4 Section 3 — Metrics Summary

Random baseline: 50.0%

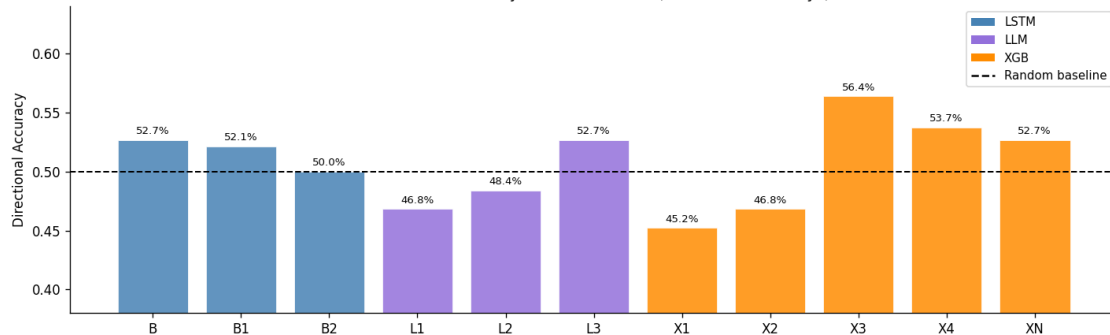
	N	Dir. Acc	MAE	RMSE
Condition				
B - LSTM regressor	188	52.7%	0.01148	0.01544
B1 - LSTM clf (single-day)	188	52.1%	-	-

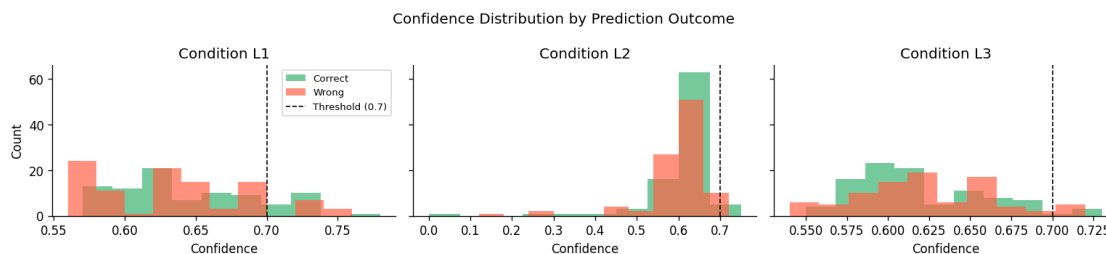
B2 - LSTM clf (sequence)	188	50.0%	-	-
L1 - LLM price only	188	46.8%	-	-
L2 - LLM news only	188	48.4%	-	-
L3 - LLM price+news	188	52.7%	-	-
X1 - XGB regressor (single-day)	188	45.2%	0.01061	0.01490
X2 - XGB regressor (20-day lags)	188	46.8%	0.01057	0.01495
X3 - XGB clf (single-day)	188	56.4%	-	-
X4 - XGB clf (20-day lags)	188	53.7%	-	-
XN - XGB clf + news	188	52.7%	-	-

Confusion Matrices — All Conditions



Directional Accuracy — All Conditions (188 evaluation days)





Condition B - Regression metrics

MAE : 0.011478

RMSE: 0.015437

(Naive baseline RMSE - std of actual returns: 0.014771)

1.5 Section 4 — Discussion

1.5.1 4.1 Summary of results

Condition	Dir. Accuracy	Pred. “up” rate
B — LSTM regressor	52.7%	47.9%
B1 — LSTM clf (single-day)	52.1%	42.0%
B2 — LSTM clf (sequence)	50.0%	51.6%
L1 — LLM price only	46.8%	63.3%
L2 — LLM news only	48.4%	68.1%
L3 — LLM price+news	52.7%	64.9%
X1 — XGB regressor (single-day)	45.2%	47.9%
X2 — XGB regressor (20-day lags)	46.8%	53.7%
X3 — XGB clf (single-day)	56.4%	72.9%
X4 — XGB clf (20-day lags)	53.7%	87.2%
XN — XGB clf + news	52.7%	84.0%

Random baseline: **50.0%**. Actual market up-rate over evaluation period: **51.6%**.

X3 records the highest directional accuracy (56.4%), but as discussed below, this is partly attributable to upward prediction bias. No condition achieves a decisive improvement over the random baseline.

1.5.2 4.2 The upward-bias problem

A recurring pattern across conditions is a systematic tendency to over-predict “up.” The effect is sharpest for XGBoost classifiers and LLM conditions:

- **LLM conditions** predict “up” on 63–68% of evaluation days, despite the market rising on only 51.6%.
- **XGB classifiers** are even more extreme: X3 predicts “up” on 72.9% of days, X4 on 87.2%, and XN on 84.0%.
- **LSTM conditions** are the exception: B predicts “up” on 47.9% (slightly contrarian), B1 on 42.0%, and B2 on 51.6% — nearly matching the actual up-rate.

For the LLM conditions, the bias plausibly reflects the bullish tone of financial text on which the model was trained. For XGBoost classifiers, the likely explanation is class imbalance within rolling training windows: during a broadly upward-trending market, “up” days dominate training observations and the classifier defaults toward the majority class. B2’s near-perfect calibration (51.6% predicted up-rate vs 51.6% actual) may reflect the sequence model learning the unconditional base rate rather than any directional signal.

The upward bias matters for interpretation: any accuracy above 51.6% can be partially achieved by predicting “up” every single day. X3’s 56.4% accuracy with a 72.9% “up” rate reflects genuine directional skill only to the degree it exceeds what naive upward bias would produce.

1.5.3 4.3 Regression vs classification framing (XGBoost)

The most striking finding for the XGBoost conditions is the large gap between regressors and classifiers:

	Single-day features	20-day lag features
Regressor (predict log return → derive direction)	X1: 45.2%	X2: 46.8%
Classifier (predict direction directly)	X3: 56.4%	X4: 53.7%

Both regressors fall **below** the random baseline, while both classifiers exceed it. The regression path introduces an extra difficulty: the model must estimate the sign of a near-zero continuous value (median absolute log return = 0.01), amplifying noise. Training the classifier directly on the binary direction target avoids this intermediate step and allows the model to focus on the boundary between up and down days.

The lag-window extension (X3 → X4) does not clearly help: X4 trails X3 by 2.7 pp. Flattening 20 days of features into a 180-dimensional vector likely introduces sparsity issues given the 250-day rolling training windows, reducing the benefit of the additional historical context.

1.5.4 4.4 Does news help?

Neither the LLM conditions nor the XGBoost classifier benefit from news.

LLM: L3 (price+news, 52.7%) outperforms L1 (price only, 46.8%) by 6 pp, but this improvement is better explained by L3’s slightly reduced upward bias (64.9% vs 63.3%) partially compensating when the combined prompt moderates the model’s bullish prior, rather than by genuine news signal extraction. L2 (news only, 48.4%) does not outperform L1, and its confidence-weighted accuracy (30.0% at 0.7 confidence, n=10) falls below random — suggesting the model is most confident precisely when it is most biased toward bullish calls on ultimately bearish days.

XGBoost: XN (X3 + PCA news embeddings, 52.7%) falls 3.7 pp short of X3 (56.4%) while simultaneously increasing the up-bias from 72.9% to 84.0%. Adding structured news embeddings

actively hurts the XGBoost classifier — the PCA-reduced vectors appear to add noise and reinforce the classifier’s majority-class tendency rather than contributing signal.

In both model families, news features as currently implemented do not add reliable directional information over price data alone.

1.5.5 4.5 LSTM classifier variants

Reframing the LSTM task from regression (B) to binary classification (B1, B2) does not yield improvement:

- **B1** (seq_len=1, single-day): 52.1% — marginally below B (52.7%), with a low predicted up-rate (42.0%), suggesting the single-timestep classifier learns a contrarian bias.
- **B2** (seq_len=20, full sequence): 50.0% — exactly random, with a predicted up-rate (51.6%) that mirrors the market’s actual rate. The sequence classifier appears to learn the unconditional base rate and little else.

The original LSTM regressor (B) outperforms both classifier variants. The directional signal in the regression target (log return magnitude and sign) may provide a richer training objective than a hard binary label at this dataset size, making the regression-then-sign approach preferable here.

1.5.6 4.6 Limitations

- **Sample size:** 188 evaluation days is a short window. A 5 pp accuracy difference corresponds to ~9 additional correct calls — well within sampling noise. No results reported here are statistically significant at conventional levels.
- **Upward-bias confounding:** for conditions with a high predicted up-rate (X3, X4, XN), directional accuracy overstates genuine skill. Precision/recall broken down by class, or a skill score corrected for base rate, would provide a cleaner comparison.
- **Non-stationarity:** the evaluation period (Jul 2025 – Apr 2026) spans varying market regimes. Performance may differ materially across sub-periods such as the tariff-driven volatility of early 2026.
- **News quality:** the Massive API descriptions are short (1–3 sentences). Richer article bodies, earnings transcripts, or SEC filings might carry stronger signal for L2, L3, and XN.
- **LLM temperature:** gpt-5-nano does not support temperature=0, so LLM predictions carry sampling variance. Running multiple passes per day and aggregating would reduce this but increases API cost.
- **LSTM capacity:** the 250-day rolling window and seq_len=20 leave limited training signal for a 2-layer LSTM. A simpler model (e.g. logistic regression on lagged returns) might perform comparably with lower variance.
- **Sentiment features unused:** the aapl_sentiments and sentiment_score columns in news_cache.csv were collected but not used in any model in this version. Incorporating Massive’s per-article AAPL sentiment as a structured feature — rather than raw text — is a natural next step.